

From Ten Benchmarks to a Q1 Biomedical Paper

Autism facial emotion recognition — what to fix in the baseline, how to structure the results, how to structure the paper, and what a reviewer will ask.

ITEM	VALUE
Corpus	2,003 images · 4 merged sources · 6 emotion classes
Class imbalance	10:1 — joy 602 vs fear 60
Current best	0.7201 accuracy · 0.6217 macro-F1 (VGG-16)
Target venues	IEEE JBHI · Computers in Biology and Medicine · AI in Medicine · BSPC
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1. The honest read

What you have built is a clean, resumable, multi-architecture benchmark with sensible training hygiene: discriminative learning rates, EMA, mixed precision, restrained augmentation, and a sampler instead of a weighted loss. That engineering is above the average submission in this space.

What you do not yet have is a result. Ten pretrained backbones fine-tuned on a merged public corpus is a benchmark report, and benchmark reports are desk-rejected at JBHI and Computers in Biology and Medicine. Worse, the specific claim you drew from the benchmark — that VGG-16 and Swin Base are the strongest architectures — is not supported by your own numbers once uncertainty is attached to them.

SEVERITY	FINDING	CONSEQUENCE
Blocker	Splits are not subject-independent	Merged face corpora contain many frames per child. Splitting at the image level puts the same face in train and test, inflating every reported metric.
Blocker	No uncertainty, so no finding	Your top three models span 0.0005 accuracy — far below what this sample size can resolve. The ranking in your table is noise presented as a result.
Fix	Two protocols reported as one	You describe a 70/15/15 split and 5-fold CV over all images. Reviewers will ask which produced the table. Pick one and describe it once.

2. The three blockers, with the arithmetic

Blocker 1 — Subject leakage

Your four sources each photograph a small number of children multiple times. Suppose a child contributes just 3 usable frames. Under a 15% image-level test split, the probability that at least one of that child's frames lands in training is 0.997. At 5 frames it is 0.9999. In practice, essentially every child in your test set was also seen during training.

The model then does not need to learn emotion. It can learn this child's face, in this lighting, in this room, and recall the label. This is the most common fatal flaw in small clinical-image papers, and JBHI reviewers look for it first.

It also silently narrows your confidence intervals. With 5 frames per child and a within-child error correlation of 0.6, your 2,003 images carry the statistical weight of roughly 589 independent samples, and honest intervals are 1.8 times wider than a naive calculation gives.

WHAT TO DO

Recover identity groups (build_manifest.py clusters faces with a FaceNet embedding), then use StratifiedGroupKFold and a group-disjoint held-out test set.

Expect your accuracy to drop. A lower, subject-independent number is publishable; a higher, leaked one is retractable.

Blocker 2 — No confidence intervals, no significance tests

Here is what your headline table looks like once uncertainty is attached. These are Wilson intervals on the numbers you already reported:

REPORTED	N	ACCURACY	95% CI	CI WIDTH
VGG-16	304	0.7201	0.667 – 0.768	±0.050
ResNet-50	304	0.6825	0.627 – 0.731	±0.052

VGG-16	2003	0.7201	0.700 – 0.739	±0.020
ResNet-50	2003	0.6825	0.662 – 0.703	±0.020

Your best-to-worst gap is 0.0376. Your best-to-third gap is 0.0005. A paired McNemar test on 2,003 samples with a typical 20% discordance rate can only detect gaps above roughly 0.028 at 80% power. So VGG-16 versus ResNet-50 is plausibly real; VGG-16 versus Swin Base versus Inception V3 is definitively not resolvable, and neither are most adjacent pairs in your ranking.

And those numbers assume independent samples, which per Blocker 1 they are not. Under cluster resampling the intervals widen further — in a dry run with realistic within-subject correlation, the cluster bootstrap interval came out 53% wider than the naive image-level one.

WHAT TO DO

Never report a bare point estimate again. Every number gets a group-level bootstrap CI. Every model comparison gets McNemar with Holm–Bonferroni correction across the ten-model family.

Then write what the statistics actually say — which is most likely that the ten backbones are statistically indistinguishable at this sample size. That is a legitimate finding, and it is the cleanest motivation you could ask for your hybrid architecture.

Blocker 3 — Merged-corpus confounding

Four datasets means four cameras, four lighting setups, four resolution and JPEG profiles, four annotation protocols. Class balance almost certainly differs across them. A model can therefore reach respectable accuracy by recognising which dataset an image came from and betting on that dataset's dominant class.

You must measure this, not assume it away. Two experiments settle it:

- Source probe. Train a classifier to predict the source dataset instead of the emotion. If balanced accuracy exceeds ~0.75, the images carry an obvious dataset fingerprint and your emotion numbers are partly artefact.
- Leave-one-dataset-out. Train on three sources, test on the fourth. This is the honest estimate of behaviour on a cohort the model has never seen, and it is what a clinical reviewer actually cares about.

There is a related labelling problem: "natural" in one dataset and "neutral" in another may not be the same construct, and FERAC has 4 classes where your scheme has 6. State the mapping explicitly, and ideally have two raters re-annotate a random 200-image subset and report Cohen's kappa.

PRIOR ART YOU NEED TO CITE

A 2025 paper in Applied Sciences used FERAC — one of your four sources — and benchmarked ResNet152V2, DenseNet201, InceptionV3, InceptionResNetV2, ViT, DeiT, CvT and Swin on it. Swin came out best at 0.800 accuracy and 0.789 F1 on four classes, and their CNNs failed entirely on the fear class — the same failure your confusion matrix shows.

That is very close to the experiment you have run, on a subset of your data. Your backbone benchmark is now the control condition, not the contribution. Search your other three sources by name before writing more.

3. The publication pipeline

Seven stages. The ordering matters: everything before Stage 4 exists so that the numbers in Stage 4 survive scrutiny. Do not start writing the paper until Stage 3 is done, because the framing of your contribution depends on what the leakage-free numbers turn out to be.

0 Provenance, ethics and licences

Before any modelling. For each of the four sources, record the licence, the original consent and IRB basis, and whether redistribution is permitted. Your merged Google Drive copy may violate at least one of them — check before the paper names it.

- Write the Ethics Statement now. JBHI and CBM both require one for human-subject image data, and children's facial images are the most scrutinised category there is.
- Decide what you can release: if not images, then the manifest of file hashes, the split assignment, and all code.

1 week · blocks everything downstream

1 Rebuild the dataset with an audit trail

Run `build_manifest.py`. It produces the manifest every later claim rests on: exact and perceptual duplicate clusters, inferred identity groups, source provenance, and a JSON audit report.

- Validate the identity clustering by eye on ~30 random clusters and report the purity you observed. Reviewers accept inferred identities when you show the validation.
- Investigate any group spanning two sources (same child in two datasets) or two labels (clustering error, or a real inter-dataset labelling conflict).
- Standardise preprocessing across sources: one face detector, one crop margin, one resize, one JPEG quality. This is your first defence against the source confound.

1–2 weeks

2 Freeze the protocol before you run it

Write down, and do not change afterwards: the CV scheme, the number of seeds, the primary metric, the held-out test set, the significance test, and the correction method. Then generate the locked test set once and never regenerate it.

- Primary metric: macro-F1, with balanced accuracy and MCC as secondaries. Plain accuracy should not lead your abstract when one class holds 30% of the data.
- Consider registering the protocol on OSF. It costs nothing and converts "we chose the best result" suspicion into a citation.

2–3 days

3 Re-run the baselines under the honest protocol

Ten backbones × 5 grouped folds × 3 seeds, plus the source probe and the LODO experiment. Save per-image probabilities for everything — every statistic afterwards is recomputed from those files, in seconds, without touching a GPU.

- Budget: roughly 150 training runs. On a single Kaggle T4 at ~12 min per run that is about 30 GPU-hours; split across sessions using the resume logic.
- Expect macro-F1 to fall. Write down the before/after gap — it becomes a figure, and it is one of the more useful things your paper can tell the field.

2–3 weeks

4 Build the contribution

The hybrid CNN–Transformer, plus the ablations that prove each component earns its place. A hybrid that beats the best baseline by 1.5 points with overlapping intervals is not a contribution; a hybrid with a mechanism motivated by ASD facial phenomenology, and an ablation showing that mechanism does the work, is.

- Ablate every component separately, at the same seed count, with the same tests.
- Report parameters, FLOPs and inference latency. Health-informatics reviewers ask whether it could run on a tablet in a clinic.

4–8 weeks

5 Interpretability and clinical framing

Grad-CAM or attention rollout, overlaid against facial action units extracted with OpenFace or Py-Feat. Show that the model attends to the AU regions the ASD literature implicates, and show where it fails to.

- Quantify it: mean attention mass inside AU-relevant regions versus outside. A picture of six heatmaps is not evidence; a number with an interval is.
- Add calibration and selective prediction. "At 80% coverage the model is 88% accurate" is the sentence a clinician can act on.

2–3 weeks

6 Write, internally review, submit

Draft against the skeleton in section 8. Then have someone outside the project read it cold and mark every claim they cannot verify from the tables.

- Fill in the CLAIM 2024 checklist and submit it as supplementary material even when not required — it pre-empts a third of typical reviewer questions.
- Prepare the code release before submission, not after acceptance.

4–6 weeks to submission

REALISTIC TIMELINE

About 4–5 months from today to a submittable manuscript, then 3–6 months of review at an Elsevier or IEEE venue. Do not compress Stages 1–3; they determine whether the paper survives review at all.

4. Where to send it

All quartiles below are SCImago 2025, verified rather than recalled. All are Q1 in Health Informatics or their primary category. Pick the primary target based on what your final contribution turns out to be, and keep the ladder in order so a rejection costs you a reformat, not a rethink.

JOURNAL	PUBLISHER	SJR	H-INDEX	FITS IF YOUR CONTRIBUTION IS...
Artificial Intelligence in Medicine	Elsevier	1.659	130	A clinically-framed method with interpretability tied to ASD phenomenology
IEEE J. Biomedical & Health Informatics	IEEE	1.624	179	A methodological advance with strong generalisation evidence. Highest bar.
Computers in Biology and Medicine	Elsevier	1.375	164	A solid applied contribution. Broadest scope, publishes a lot.
Biomedical Signal Processing & Control	Elsevier	1.336	141	An architecture paper with clinical application. Most forgiving of the four.
J. Autism & Developmental Disorders	Springer	1.169	234	A findings-first paper about ASD expression itself, not the architecture

TWO CAUTIONS

On Computers in Biology and Medicine: it has grown very large (1,750 documents in 2025). Some institutions have become wary of high-volume Q1 venues. Check your university's list before committing.

On dataset size: 2,003 images across an unknown but small number of children is modest for any of these. Do not oversell. Frame the paper as a rigorous benchmark plus a motivated architecture on a genuinely scarce clinical population, and let methodological rigour carry it rather than headline accuracy.

5. What to check before you trust a single number

Work through this in order. Anything marked Blocker invalidates the results underneath it, so there is no point running the next stage until it is cleared.

#	CHECK	SEVERITY	HOW YOU KNOW IT PASSED
1	Exact duplicates removed	Blocker	MD5 collision count is zero after cleaning
2	Near-duplicates clustered	Blocker	pHash Hamming ≤ 6 clusters identified; one representative kept, no cluster straddles a fold
3	Subject-independent splits	Blocker	Assertion in code: train and test group sets are disjoint, per fold
4	Locked test set	Blocker	Group list written to disk once; evaluated exactly once, at the end
5	Label mapping documented	Blocker	Explicit table of source label $\rightarrow$ canonical class, in the paper
6	Multi-seed repetition	Fix	≥ 3 seeds; seed SD reported alongside every mean
7	Group-level bootstrap CIs	Fix	Every headline metric carries a 95% interval, resampled over subjects
8	Paired significance tests	Fix	McNemar per pair, Holm-corrected; conclusions match the corrected p-values
9	Source probe run	Fix	Balanced accuracy of dataset-ID prediction reported and discussed
10	Leave-one-dataset-out	Fix	A LODO table exists; the drop from in-corpus performance is stated plainly
11	Per-class metrics with support	Fix	Every per-class F1 shows its n; classes with $n < 30$ flagged as unstable
12	Calibration	Strengthen	ECE, Brier, reliability curve; temperature fitted on validation only
13	Selective prediction	Strengthen	Risk–coverage curve and AUROC; accuracy at 80% coverage quoted
14	Subgroup analysis	Strengthen	Performance by source, and by age/sex/skin tone where recoverable
15	Interpretability	Strengthen	Grad-CAM quantified against AU regions, not just displayed
16	Efficiency reported	Strengthen	Params, GFLOPs, latency on a named device
17	Reproducibility	Strengthen	Seeds, deterministic flags, library versions, git commit, config hash logged

Two more things specific to a biomedical venue

- An ASD-versus-typical contrast. Your entire motivation is that ASD faces are harder. Prove it: fine-tune the same backbone on a typically-developing children's FER set (CAFE, or a child subset of RAF-DB or AffectNet) under an identical protocol and report the gap. Without this your premise is an assertion. With it, you have a headline finding.
- A clinically meaningful operating point. Accuracy is not a clinical quantity. Pick a use case — screening support, therapy-session feedback — state the cost asymmetry, and report sensitivity at a fixed specificity, or accuracy at a fixed coverage. This is what moves a paper from "CV paper submitted to a medical journal" to "medical informatics paper".

6. Implementing the baseline, step by step

Step 1 — Build the manifest

Everything downstream reads one CSV. Nothing reads the directory tree directly, so the split logic cannot accidentally see a file the audit did not. Read `manifest_audit.json` before going further — a non-zero `groups_spanning_labels` means either a clustering error or a genuine labelling conflict between sources.

Step 2 — Standardise preprocessing across sources

Run one face detector (MTCNN or RetinaFace) over everything, with the same crop margin, the same output size, and the same encoder settings. Different resolutions and compression profiles are exactly the fingerprint the source probe will find.

Step 3 — Carve the locked test set, by group

Roughly 15% of images, split at the group level, stratified as closely as class proportions allow. Written to `locked_test_groups.json` and reused on every subsequent run. You touch it once, in the final table.

Step 4 — Grouped cross-validation on the remainder

StratifiedGroupKFold with five splits, with an assertion that no group appears on both sides of a fold. Repeat over three seeds. The seed loop is not optional: with ~1,700 development images, run-to-run variation is comparable to the differences between your architectures.

Step 5 — Train with the hygiene you already have, plus two fixes

Keep the sampler and the unweighted loss — doing both is double-counting, and you already avoid it, so say so explicitly because reviewers check. Fix one: select on macro-F1, not accuracy, because accuracy-based early stopping quietly selects the checkpoint that abandoned fear. Fix two: reconsider horizontal flipping, which imposes a symmetry prior that works against your stated asymmetry hypothesis.

Step 6 — Save predictions, not just metrics

Per-image probabilities, labels, groups, sources and paths go to an `.npz` per model per seed. This is the highest-leverage habit in this whole document: every statistic, table and figure is then recomputed in seconds from disk, and when a reviewer asks for a metric you did not think of, you produce it the same day instead of re-running 150 GPU jobs.

Step 7 — Run the confound experiments

Source probe and LODO. Both are cheap relative to the main benchmark, and both directly answer the question a health-informatics reviewer will ask about a four-way merged corpus.

Step 8 — Compute the statistics from the saved predictions

Cluster bootstrap CIs, McNemar with Holm correction, per-class metrics with support, calibration, risk-coverage, subgroup breakdown. All deterministic, all reproducible from files you can attach to the submission.

7. How to organise the Results section

Nine subsections in this order. The logic is a funnel: establish that the data are trustworthy, then that the benchmark is honest, then what generalises, then where it fails, then why your method helps.

§	SUBSECTION	WHAT IT CONTAINS	ARTEFACTS
4.1	Corpus characterisation and leakage audit	The unified corpus, duplicate and identity audit, class × source matrix. Earns the reader's trust before showing a single accuracy.	Table 1; Figure 1 (dataset flow)
4.2	Baseline benchmark, subject-independent CV	Ten backbones, macro-F1 first, every value with a group-bootstrap CI and seed SD. Do not bold the top row until §4.3 says you may.	Table 2; Figure 2 (forest plot)
4.3	Statistical comparison of baselines	McNemar against the top model, Holm-corrected. State plainly how many differences survive correction.	Table 3; optional critical-difference diagram
4.4	Generalisation: LODO and source probe	Honest cross-cohort numbers and the measured strength of the dataset fingerprint. Put in-corpus and LODO side by side.	Table 4; probe accuracy quoted inline
4.5	Minority-class behaviour	Per-class precision, recall and F1 with support, plus the row-normalised confusion matrix. Fear has ~60 images total; report the interval.	Table 5; Figure 3 (confusion matrix)
4.6	Calibration and selective prediction	ECE, Brier, reliability, risk-coverage, AUROC. Then the clinical sentence about abstention.	Table 6; Figure 4
4.7	Proposed architecture and ablations	Your hybrid against the best baseline with the same intervals and paired tests, then one row per removed component.	Tables 7 and 8
4.8	Interpretability	Grad-CAM on correct and incorrect predictions, overlaid with AU regions, plus the quantitative overlap measure. Include failure cases.	Figure 5; Table 9
4.9	Computational cost	Parameters, GFLOPs, training time, inference latency on a named device. Short, but its absence is a standard complaint.	Table 10

Rules that apply to every subsection

- Never a bare point estimate. Number, interval, and the n it rests on.
- Never claim a difference you have not tested. "Highest" is descriptive and fine; "outperforms" is a statistical claim and needs a p-value.
- Results describe, Discussion interprets. Keep "this suggests CNNs capture complementary features" out of section 4 entirely.
- Every figure must survive greyscale printing and be legible at single-column width. Vector where possible, 300 dpi where not.
- Drop the radar chart. Radar plots encode area, which is not meaningful across metrics on different scales, and they hide uncertainty. A forest plot of macro-F1 with intervals carries strictly more information. Same for the model correlation heatmap: if it motivates the hybrid design it belongs in the Discussion, not among your results.

8. Full paper skeleton

Target length for JBHI or CBM: 8,000–10,000 words, 8–10 figures and tables combined. Word counts are guides for a first draft, not limits.

SECTION	WORDS	WHAT MUST BE IN IT
Title	~15	Name the population, the task and the mechanism. Not "A Deep Learning Approach for...". Closer to: Asymmetry-aware CNN–Transformer fusion for facial emotion recognition in children with autism: a subject-independent multi-corpus benchmark.
Abstract	200–250	Structured if allowed. Background → the specific gap → data (n images, n subjects, n sources) → method → primary result with CI → the LODO number → one clinical implication. Lead with macro-F1, not accuracy.
1. Introduction	900–1200	Why ASD emotion recognition matters clinically. Why ASD faces differ — cite the phenomenology, do not assert it. Why existing FER models fail here. Then three or four numbered contributions, each checkable against a specific table.
2. Related Work	1000–1400	Four threads: FER on typical faces; FER in ASD specifically; CNN–Transformer hybrids; and — the one most papers omit — evaluation methodology on small clinical image corpora. That fourth thread is how you set up your rigour as a contribution.
3.1 Datasets	700–900	Each source: origin, licence, consent basis, n, class scheme. The label mapping table. Deduplication and identity grouping with validation. The class × source matrix. Be exhaustive; it buys credibility everywhere else.
3.2 Preprocessing	300–400	Face detection, alignment, crop margin, resize, encoding — identical across sources. State explicitly that this was harmonised, and why.
3.3 Evaluation protocol	500–700	The whole protocol in one place: subject-independent grouping, StratifiedGroupKFold, seeds, the locked test set, primary and secondary metrics, the bootstrap procedure and its resampling unit, the significance tests and the correction. Written so someone could reimplement it.
3.4 Baselines	400–500	The ten backbones and why those ten. Training configuration. The sampler-not-loss decision, stated as a decision. Augmentation policy and why it is restrained.
3.5 Proposed method	900–1300	Architecture diagram, the mechanism, and the clinical motivation for each component. "We added attention" is not a contribution. "ASD facial expression shows reduced hemifacial synchrony, so we compute features on the two hemifaces separately and fuse them through cross-attention" is.
4. Results	1800–2400	The nine subsections in section 7. Description only.
5. Discussion	1200–1600	What the results mean clinically. Why fear and surprise confuse (they share AU1+AU2+AU5; say so). How your interpretability maps onto the ASD literature. Comparison to published FER numbers with an explicit warning that most used image-level splits and are not comparable — this turns your lower number into evidence of rigour.
6. Limitations	350–500	Its own section, not a paragraph in the Discussion. Small n; subject count unknown or small; identities inferred rather than given; no independent prospective cohort; posed rather than spontaneous expressions; unknown demographic composition; no clinical validation.
7. Conclusion	200–300	What was shown, what it enables, what comes next. No new claims.
Declarations	—	Ethics and consent, data availability, code availability, funding, CRediT contributions, conflicts of interest, and any AI-assistance disclosure the journal requires.
Supplementary	—	CLAIM 2024 checklist, full hyperparameter tables, per-fold results, additional confusion

matrices, the identity-clustering validation, extra Grad-CAM panels.

9. Where the novelty actually comes from

"CNN + Transformer hybrid" is, on its own, a 2021 contribution. You need a mechanism a reviewer can trace back to something true about autistic facial expression. Your own project description already contains the strongest candidate.

The asymmetry angle — take this one

You wrote that children with ASD often express emotions with less symmetrical or synchronised facial movements. That is a testable architectural prior, and almost nobody has built on it:

- Split the aligned face into left and right hemifaces. Encode each with a shared-weight CNN. Fuse through cross-attention between the two streams, so the model represents hemiface disagreement explicitly rather than averaging it away.
- Add an asymmetry-magnitude head as an auxiliary task, or an asymmetry-derived attention gate.
- The ablation writes itself: symmetric baseline vs hemiface-split vs hemiface-split with cross-attention.
- It produces a genuinely interesting secondary result — a measured asymmetry signature per emotion class, which a clinical readership cares about independently of your accuracy.

It also resolves the flip-augmentation tension in your favour: with a hemiface architecture, disabling horizontal flip is principled rather than arbitrary.

Three weaker alternatives

APPROACH	ASSESSMENT
AU-guided attention	Extract action units with OpenFace, use them to supervise or gate attention. Interpretable by construction. Solid, but well-trodden.
Domain generalisation across sources	Treat each dataset as a domain; apply DANN, CORAL or similar; show LODO improvement. Respectable and directly targets your confound, but the novelty is applied rather than architectural.
Few-shot / metric learning for rare classes	Prototypical networks or supervised contrastive learning for fear and surprise. Fits the data scarcity honestly, though the tiny n makes evaluation fragile.

A FRAMING THAT WORKS EVEN IF THE HYBRID DISAPPOINTS

If your architecture ends up statistically tied with the best baseline, you still have a paper — a different one. Retitle it as a rigorous subject-independent benchmark that quantifies how much published ASD-FER performance is attributable to subject leakage and dataset confounding, and releases the audited protocol.

That is a genuine service to a field where nearly every paper splits at the image level, and CBM and BSPC both publish that kind of work. Decide which paper you are writing after Stage 3, not before.

10. Objections you will get, and what to have ready

THE OBJECTION	WHAT YOU NEED ALREADY IN THE MANUSCRIPT
"Same subjects appear in train and test."	§3.3 describes identity grouping and StratifiedGroupKFold; §3.1 reports the clustering validation; the code asserts group disjointness per fold.
"Differences between models are within noise."	Table 2 carries CIs, Table 3 carries Holm-corrected McNemar, and your prose already concedes which comparisons are unresolvable.
"Accuracy is inflated by merging four datasets."	The source-probe number and the LODO table in §4.4, with the drop stated plainly rather than buried.

"Only 2,003 images. This is underpowered."	A limitations section that says so first, plus a power statement giving the minimum difference your design can detect. Owning it is the strong move.
"Fine-tuning ten pretrained backbones is not novel."	The novelty is §3.5 plus the audited protocol; the benchmark is the control condition, and the Introduction must say that explicitly.
"Why is this a medical paper?"	Calibration, selective prediction at a clinical operating point, AU-grounded interpretability, subgroup fairness, and the ASD-versus-typical contrast.
"Labels come from four different annotation protocols."	The label-mapping table and, ideally, a two-rater re-annotation of a 200-image subset with Cohen's kappa.
"How does this compare to published ASD-FER work?"	A comparison table that includes each prior work's split protocol as a column, with a note that image-level-split numbers are not comparable to yours.
"Was ethics approval obtained?"	Per-source consent and IRB provenance in §3.1 and the declarations. For children's facial images this will be asked.
"Can this run in a clinical setting?"	The efficiency table, and one paragraph in the Discussion on the deployment scenario you have in mind.

11. Pre-submission checklist

Do not submit while any of these is unticked.

Science

- Every headline number carries a group-level 95% CI and its n
- Every comparative claim maps to a corrected p-value in a table
- The locked test set was evaluated exactly once, and the manuscript says so
- LODO and source-probe results are reported, including if unflattering
- Per-class metrics show support; classes with $n < 30$ are flagged
- Ablations cover every claimed component, at the same seed count
- Limitations names small n, inferred identities, and the absence of a prospective cohort

Reporting

- CLAIM 2024 checklist completed and attached
- Ethics, consent and data-availability statements written and accurate
- Per-source licences verified as permitting your use and any redistribution
- Code repository prepared, with seeds, versions, config and the split manifest
- Figures legible in greyscale at single-column width, 300 dpi or vector
- Journal formatting, reference style and word limits applied
- Cover letter names the specific gap and the specific contribution in three sentences

Prudence

- One person outside the project has read it cold and challenged every claim
- Reviewer suggestions prepared (most Elsevier journals request them)
- A second- and third-choice journal already chosen, in the section 4 order

THE ONE SENTENCE TO KEEP IN VIEW

A modest, leakage-free, well-quantified result published in a Q1 journal is worth considerably more than an impressive number that a reader cannot reproduce. Everything above is in service of the first thing.

Journal quartiles, SJR values and h-indices are SCImago 2025 figures, retrieved 30 August 2026. Reporting-guideline reference: Checklist for Artificial Intelligence in Medical Imaging (CLAIM), 2024 update, Radiology: Artificial Intelligence. Confidence intervals and power figures in section 2 were computed from the counts in the project progress report. Prior-art reference: Applied Sciences 15(17):9555, 2025.